

JavaScript ES6 Challenge #2 (Bonus)

Scinario

لإدارة الموظفين Dashboard أنت شغال في شركة بتعمل.

عندك الداتا دي:

```
``js
```

```
const employees = [
```

```
{
```

```
  id: 1,
```

```
  name: "Ahmed",
```

```
  age: 22,
```

```
  salary: 6000,
```

```
  department: "IT",
```

```
  active: true
```

```
},
```

```
{
```

```
  id: 2,
```

```
  name: "Sara",
```

```
  age: 27,
```

```
  salary: 8500,
```

```
  department: "HR",
```

```
    active: true
  },
  {
    id: 3,
    name: "Ali",
    age: 20,
    salary: 4500,
    department: "IT",
    active: false
  },
  {
    id: 4,
    name: "Mona",
    age: 30,
    salary: 10000,
    department: "Finance",
    active: true
  },
  {
    id: 5,
```

```
    name: "Omar",
    age: 24,
    salary: 7000,
    department: "Marketing",
    active: false
},
{
    id: 6,
    name: "Youssef",
    age: 29,
    salary: 12000,
    department: "IT",
    active: true
}
];
...
```

Part 1 - Loops

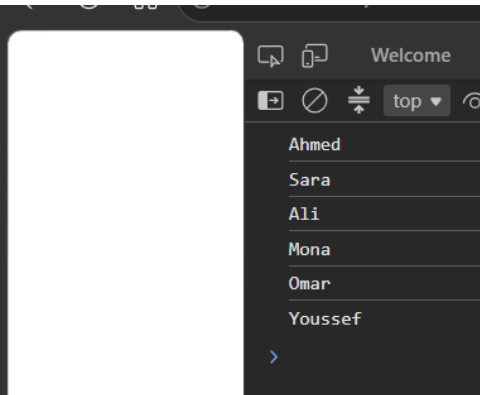
المطلوب

1

اطبع أسماء كل الموظفين باستخدام

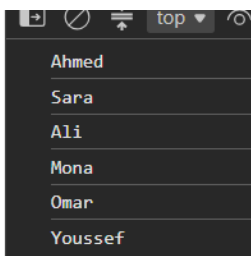
- for

```
2 const employees = [
3   { id: 1, name: "Ahmed", department: "IT", salary: 6000, a
4   { id: 2, name: "Sara", department: "HR", salary: 8500, a
5   { id: 3, name: "Ali", department: "IT", salary: 4500, ad
6   { id: 4, name: "Mona", department: "Finance", salary: 12
7   { id: 5, name: "Omar", department: "IT", salary: 7000, a
8   { id: 6, name: "Youssef", department: "HR", salary: 5000
9 ];
10
11 for (let i = 0; i < employees.length; i++) {
12   console.log(employees[i].name);
13 }
14
```



- for...of

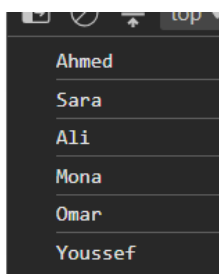
```
8   { id: 6, name: "Youssef", department: "HR", salary: 5000
9 ];
10
11 for (let emp of employees) {
12   console.log(emp.name);
13 }
14
```



- forEach

```
   { id: 6, name: "Youssef", department: "HR", salary: 5000
];

employees.forEach(emp => console.log(emp.name));
```



2

for...in باستخدام Index اطبع الـ

```
{ id: 6, name: Youssef , department:
};

for (let index in employees) {
  console.log(index);
}
```



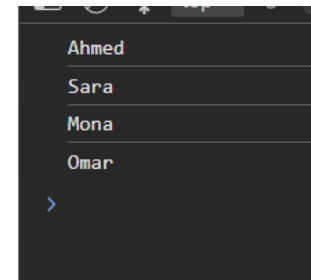
0
1
2
3
4
5

3

عادية Loop فقط باستخدام (active = true) اطبع الموظفين النشطين.

```
};

for (let i = 0; i < employees.length; i++) {
  if (employees[i].active) {
    console.log(employees[i].name);
  }
}
```



Ahmed
Sara
Mona
Omar
>

Part 2 - ES6

المطلوب

1

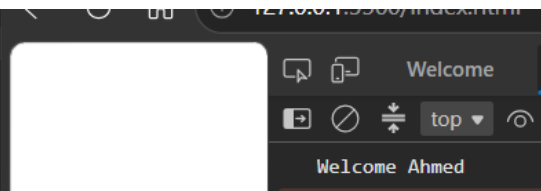
Arrow Function حوّل الدالة دي لـ

```
```js
```

```
function welcome(name){
 return "Welcome " + name;}

```

```
1
2 const welcome = (name) => `Welcome ${name}`;
3 console.log(welcome("Ahmed"));
4 console.log(welcome(employees[0].name));
```



### 2

استخدم Destructuring

```
```js
```

```
const employee = employees[0];
```

```
```
```

واطلع

- name

- salary

```

```

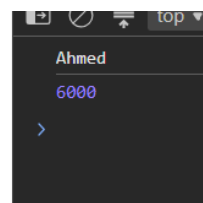
```
const employee = employees[0];
```

```
const {name, salary} = employee
```

```
const employee = employees[0];

const {name, salary} = employee;

console.log(name);
console.log(salary);
```



### 3

وزود فيها Spread Operator باستخدام employee اعمل نسخة جديدة من الـ

```
```js
```

```
country:"Egypt"
```

```
const employee = employees[0];
const newEmployee = {...employee, country: "Egypt" };
console.log(newEmployee);
```

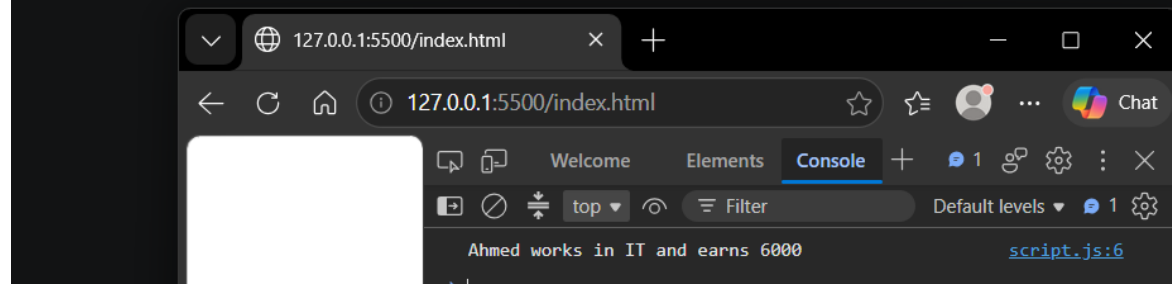
```
script.js:8
  {name: 'Ahmed', salary: 6000, department: 'IT', country:
    'Egypt'}
```

4

واطبع Template Literal استخدم

Ahmed works in IT and earns 6000

```
1 const employees = [
2   { name: "Ahmed", salary: 6000, department: "IT" }
3 ];
4
5 const employee = employees[0];
6 console.log(`${employee.name} works in ${employee.department} and earns ${employee.salary}`);
```

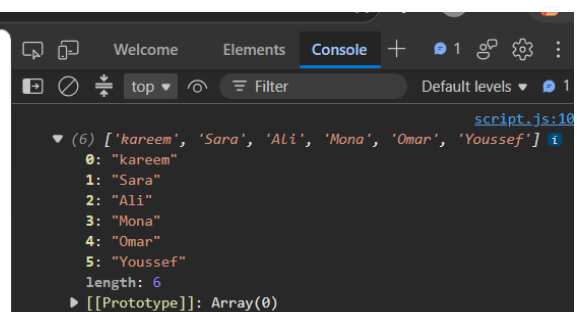


Part 3 - map()

1

فيها أسماء الموظفين بس Array اعمل

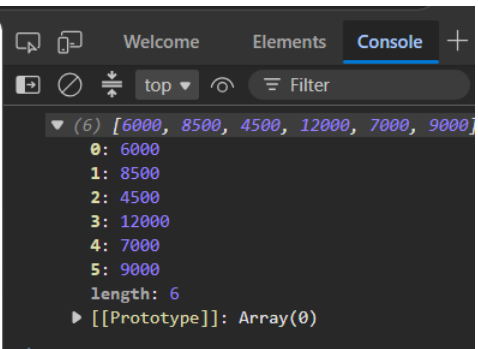
```
const employees = [
  { name: "kareem", salary: 6000 },
  { name: "Sara", salary: 8500 },
  { name: "Ali", salary: 4500 },
  { name: "Mona", salary: 12000 },
  { name: "Omar", salary: 7000 },
  { name: "Youssef", salary: 9000 }
];
const names = employees.map(emp => emp.name);
console.log(names);
```



2

فيها الرواتب فقط Array اعمل

```
const employees = [
  { name: "kareem", salary: 6000 },
  { name: "Sara", salary: 8500 },
  { name: "Ali", salary: 4500 },
  { name: "Mona", salary: 12000 },
  { name: "Omar", salary: 7000 },
  { name: "Youssef", salary: 9000 }
];
const salaries = employees.map(emp => emp.salary);
console.log(salaries);
```



3

بالشكل ده Array اعمل

```
``js
```

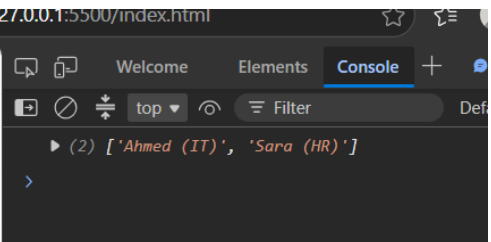
```
[
```

```
"Ahmed (IT)",
```

```
"Sara (HR)"
```

```
]
```

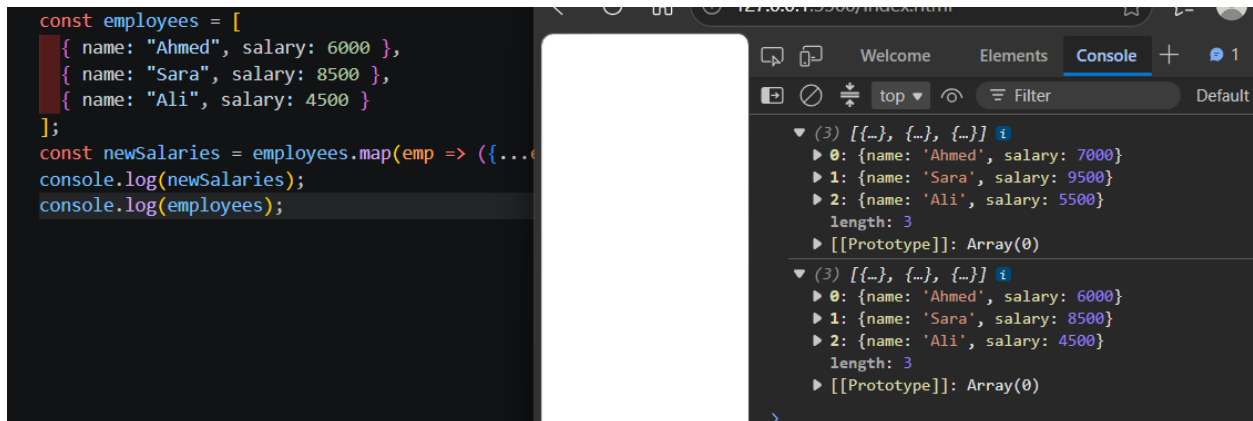
```
const result = [
  "Ahmed (IT)",
  "Sara (HR)"
];
console.log(result);
```



4

جديدة تزود مرتب كل موظف 1000 جنيه Array اعمل

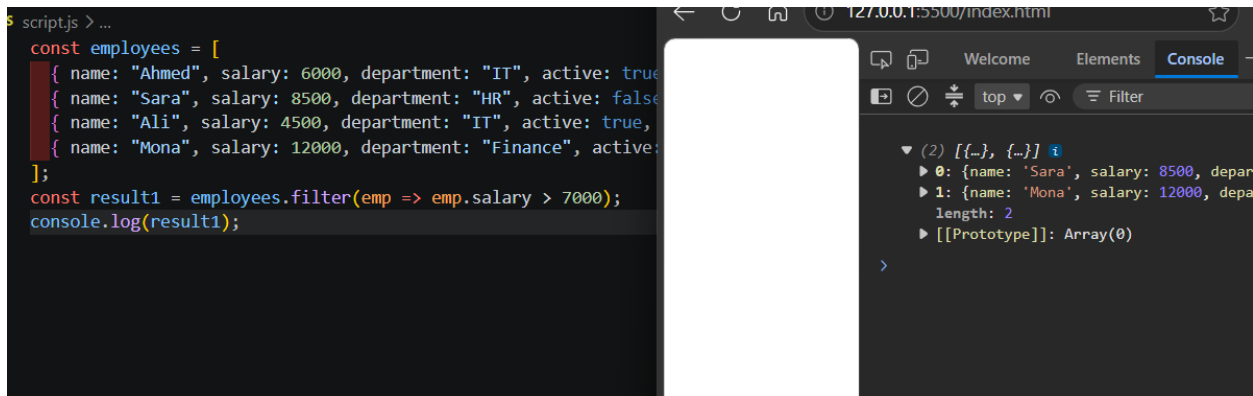
> الأصلية Array بدون تعديل الـ



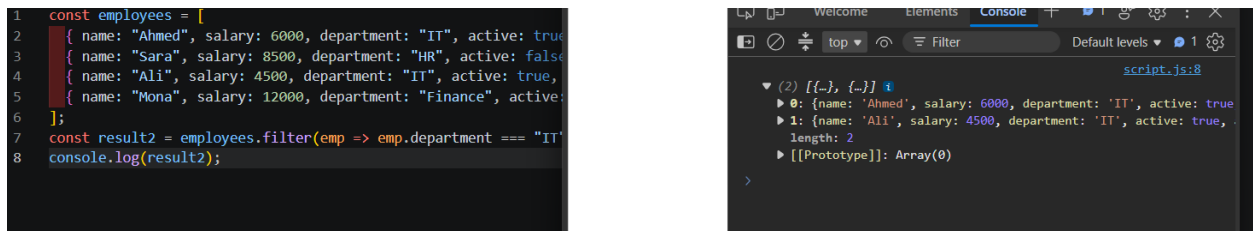
Part 4 - filter()

طلع

1. الموظفين اللي مرتباتهم أكبر من 7000.

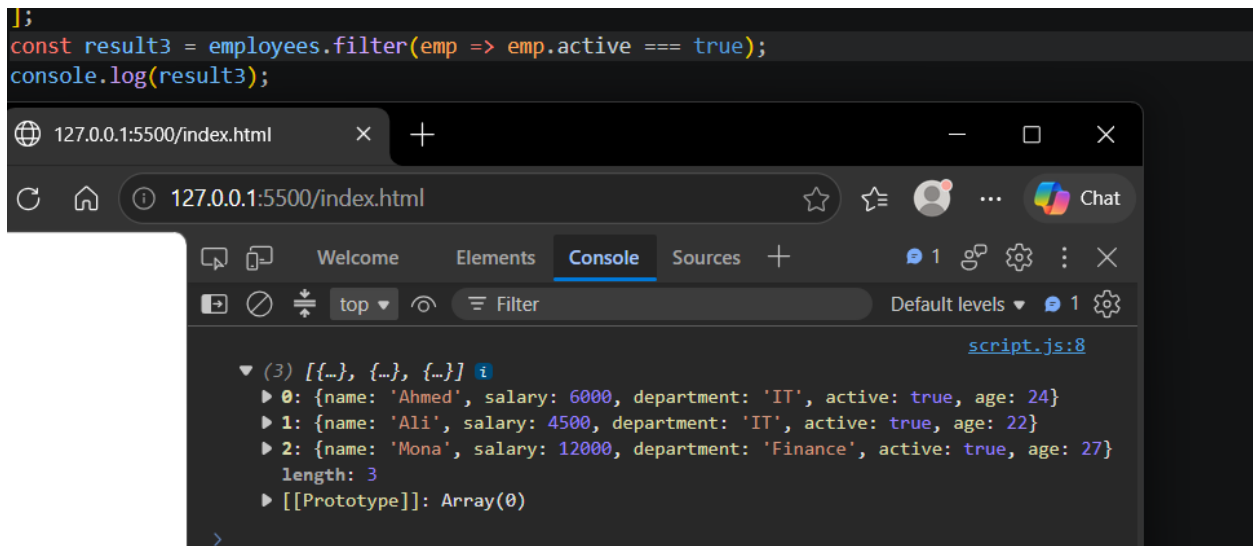


2. IT الموظفين اللي في قسم.



3. الموظفين النشطين.

```
};
const result3 = employees.filter(emp => emp.active === true);
console.log(result3);
```



127.0.0.1:5500/index.html

127.0.0.1:5500/index.html

script.js:8

(3) [{...}, {...}, {...}]

- 0: {name: 'Ahmed', salary: 6000, department: 'IT', active: true, age: 24}
- 1: {name: 'Ali', salary: 4500, department: 'IT', active: true, age: 22}
- 2: {name: 'Mona', salary: 12000, department: 'Finance', active: true, age: 27}

length: 3

[[Prototype]]: Array(0)

4. الموظفين سنهم أقل من 25.

```
const result4 = employees.filter(emp => emp.age < 25);
console.log(result4);
```



script.js:8

(2) [{...}, {...}]

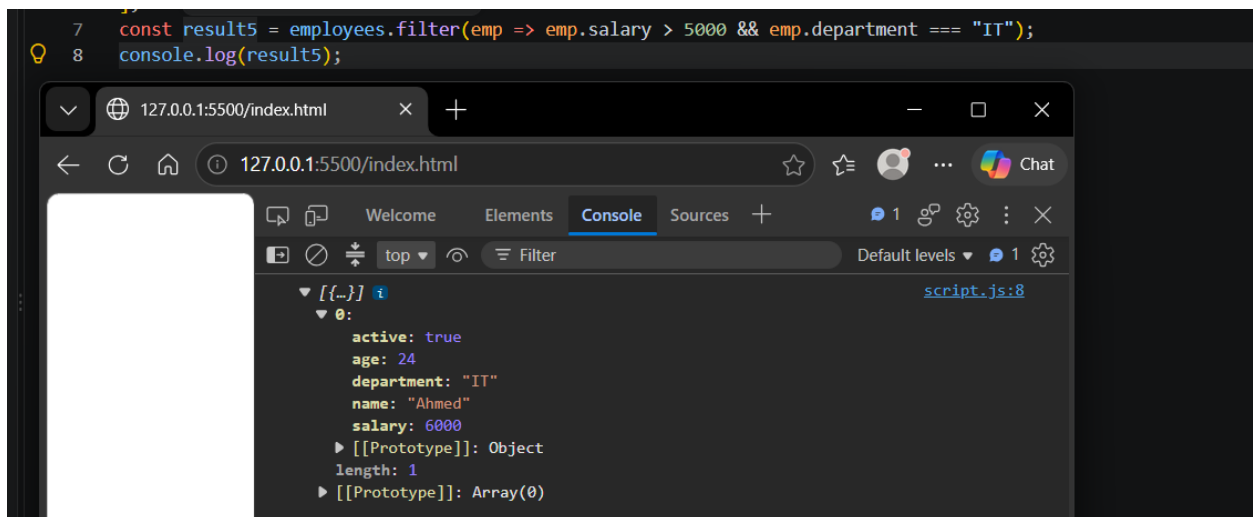
- 0: {name: 'Ahmed', salary: 6000, department: 'IT', active: true, age: 24}
- 1: {name: 'Ali', salary: 4500, department: 'IT', active: true, age: 22}

length: 2

[[Prototype]]: Array(0)

5. ومرتبهم أكبر من 5000 IT الموظفين اللي في قسم.

```
const result5 = employees.filter(emp => emp.salary > 5000 && emp.department === "IT");
console.log(result5);
```



script.js:8

(1) [{...}]

- 0: {active: true, age: 24, department: "IT", name: "Ahmed", salary: 6000}

[[Prototype]]: Object

length: 1

[[Prototype]]: Array(0)

Part 5 - find()

هات ###

1. أول موظف مرتبه أكبر من 9000

```
const employees = [
  { id: 1, name: "Ahmed", salary: 6000, department: "IT", active: true },
  { id: 2, name: "Sara", salary: 8500, department: "HR", active: false },
  { id: 3, name: "Ali", salary: 12000, department: "IT", active: true },
  { id: 4, name: "Mona", salary: 9500, department: "HR", active: true }
];
const result1 = employees.find(emp => emp.salary > 9000);
console.log(result1);
```

```
▼ {id: 3, name: 'Ali', salary: 12000, department: 'IT', active: true}
  active: true
  department: "IT"
  id: 3
  name: "Ali"
  salary: 12000
  ► [[Prototype]]: Object
```

2. أول موظف في قسم HR.

```
{ id: 2, name: "Sara", salary: 8500, department: "HR", active: false },
{ id: 3, name: "Ali", salary: 12000, department: "IT", active: true },
{ id: 4, name: "Mona", salary: 9500, department: "HR", active: true }
];
const firstHR = employees.find(emp => emp.department === "HR");
console.log(firstHR.name);
```

```
Sara
>
```

3. أول موظف غير نشط.

```
{ id: 2, name: "Sara", salary: 8500, department: "HR", active: false },
{ id: 3, name: "Ali", salary: 12000, department: "IT", active: true },
{ id: 4, name: "Mona", salary: 9500, department: "HR", active: true }
];
const result3 = employees.find(emp => emp.unactive === true);
console.log(result3.name);
```

```
Ali
> |
```

4. بتاعه id 100 جرب تجيب موظف الـ.

```
{ id: 3, name: "Ali", salary: 12000, department: "IT", unactive: true },
{ id: 100, name: "Mona", salary: 9500, department: "HR", active: true }
];
const result4 = employees.find(emp => emp.id === 100);
console.log(result4.name);
```

```
Mona
> |
```

Part 6 - Mixed Challenge

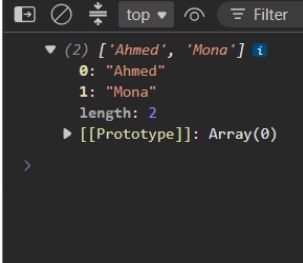
المطلوب

عادية Loop اعمل النتائج دي بدون استخدام

1

فيها أسماء الموظفين النشطين فقط Array اعمل

```
{ id: 1, name: "Ahmed", salary: 6000, department: "IT", active: true },
{ id: 2, name: "Sara", salary: 8500, department: "HR", active: false },
{ id: 3, name: "Ali", salary: 12000, department: "IT", unactive: true },
{ id: 100, name: "Mona", salary: 9500, department: "HR", active: true }
];
const activeNames = employees
  .filter(emp => emp.active)
  .map(emp => emp.name);
console.log(activeNames);
```

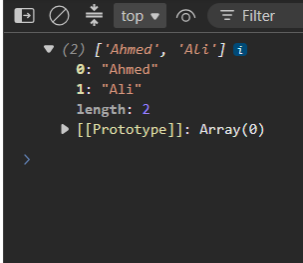


```
(2) ['Ahmed', 'Mona']
0: "Ahmed"
1: "Mona"
length: 2
[[Prototype]]: Array(0)
```

2

فقط IT فيها أسماء موظفين الـ Array اعمل

```
{ id: 1, name: "Ahmed", salary: 6000, department: "IT", active: true },
{ id: 2, name: "Sara", salary: 8500, department: "HR", active: false },
{ id: 3, name: "Ali", salary: 12000, department: "IT", unactive: true },
{ id: 100, name: "Mona", salary: 9500, department: "HR", active: true }
];
const itNames = employees
  .filter(emp => emp.department === "IT")
  .map(emp => emp.name);
console.log(itNames);
```



```
(2) ['Ahmed', 'Ali']
0: "Ahmed"
1: "Ali"
length: 2
[[Prototype]]: Array(0)
```

3

فيها أسماء الموظفين اللي مرتبهم أكبر من 7000 Array اعمل

```

const employees = [
  { id: 1, name: "Ahmed", salary: 6000, department: "IT", active: true },
  { id: 2, name: "Sara", salary: 8500, department: "HR", active: false },
  { id: 3, name: "Ali", salary: 12000, department: "IT", unactive: true },
  { id: 100, name: "Mona", salary: 9500, department: "HR", active: true }
];

const highSalaryNames = employees
  .filter(emp => emp.salary > 7000)
  .map(emp => emp.name);

console.log(highSalaryNames);

```

```

(3) ['Sara', 'Ali', 'Mona']
  0: "Sara"
  1: "Ali"
  2: "Mona"
  length: 3
  [[Prototype]]: Array(0)

```

4

بالشكل ده Array اعمل

```

```js

```

```

[

```

```

{

```

```

 employee:"Ahmed",

```

```

 bonus:600

```

```

},

```

```

...

```

```

]

```

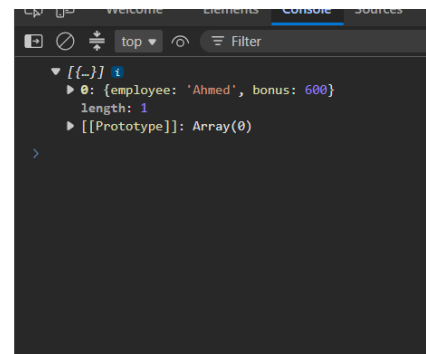
```

```

```

.البونص = 10% من المرتب >

```
const employees = [
  { name: "Ahmed", salary: 6000 },
  { name: "Sara", salary: 8500 },
  { name: "Ali", salary: 4500 }
];
const ahmedBonus = employees
  .filter(emp => emp.name === "Ahmed")
  .map(emp => ({
    employee: emp.name,
    bonus: emp.salary * 0.10
  }));
console.log(ahmedBonus);
```



5

فيها أول حرف من اسم كل موظف Array اعمل

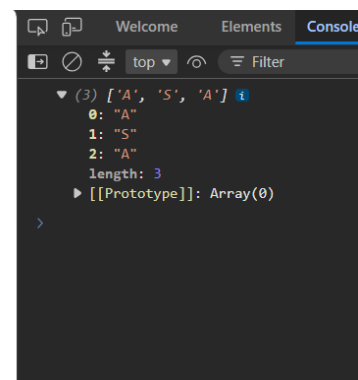
مثال

```
```js
```

```
["A","S","A","M"]
```

```
```
```

```
const employees = [
  { name: "Ahmed", salary: 6000 },
  { name: "Sara", salary: 8500 },
  { name: "Ali", salary: 4500 }
];
const firstLetters = employees.map(emp => emp.name[0]);
console.log(firstLetters);
```



Part 7 - Logical Thninking

عندك

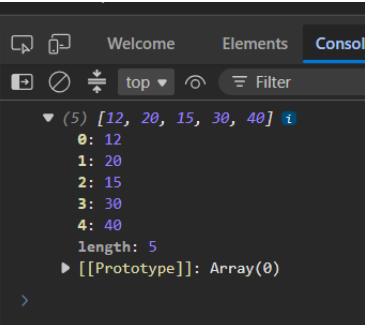
```js

const numbers = [5,12,8,20,15,30,3,40];

### المطلوب

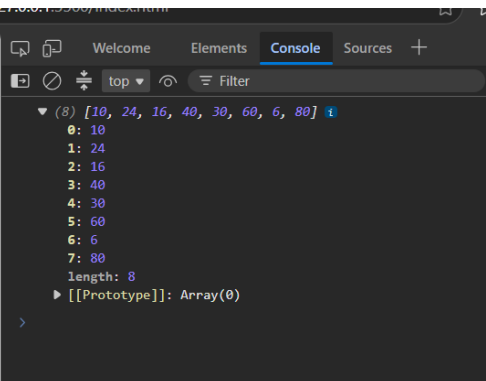
1. طلع الأرقام الأكبر من 10.

```
js > JS script.js > ...
1 const numbers = [5,12,8,20,15,30,3,40];
2 const biggerThan10 = numbers.filter(n => n > 10);
3 console.log(biggerThan10);
```



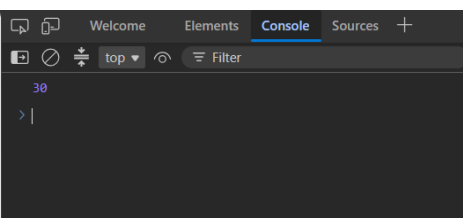
2. اضرب كل رقم  $2 \times$ .

```
js > JS script.js > ...
1 const numbers = [5,12,8,20,15,30,3,40];
2 const doubled = numbers.map(n => n * 2);
3 console.log(doubled);
```



3. هات أول رقم أكبر من 25.

```
js > JS script.js > ...
1 const numbers = [5,12,8,20,15,30,3,40];
2 const firstBiggerThan25 = numbers.find(n => n > 25);
3 console.log(firstBiggerThan25);
```

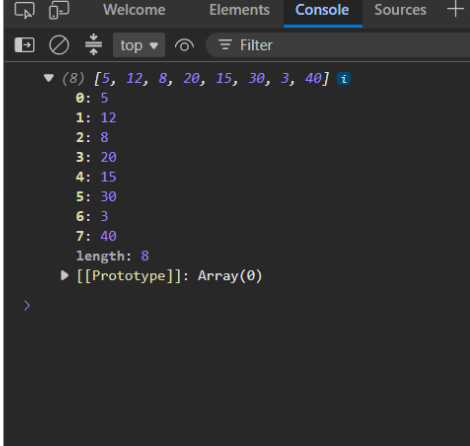


4. اطبع كل الأرقام.

```

1 const numbers = [5,12,8,20,15,30,3,40];
2 console.log(numbers);

```



بالشكل String فيها Array اعمل 5.

...

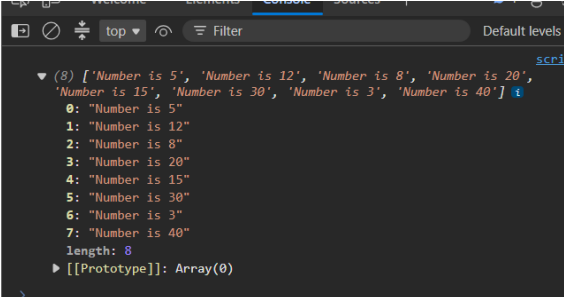
Number is 5

Number is 12

```

const numbers = [5,12,8,20,15,30,3,40];
const stringsArray = numbers.map(n => `Number is ${n}`);
console.log(stringsArray);

```



## # Part 8 - Object Challenge

عندك

```
```js
```

```
const product = {
```

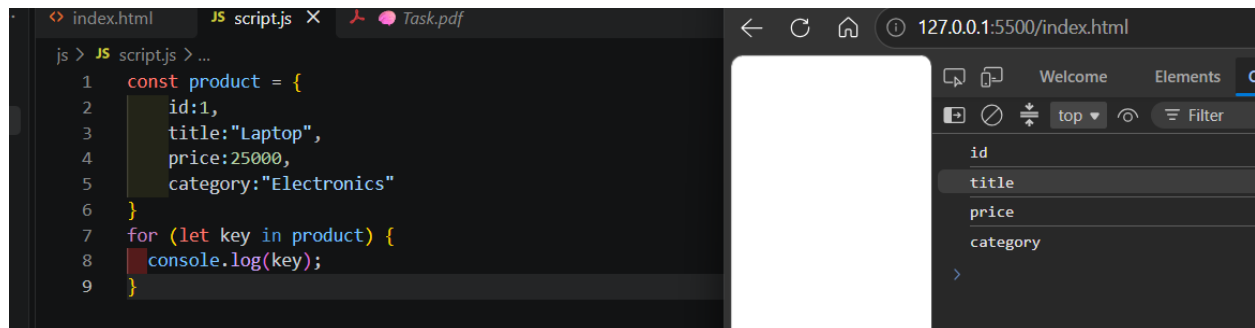
```
  id:1,
```

```
  title:"Laptop",
```

```
    price:25000,  
    category:"Electronics"  
  }  
}
```

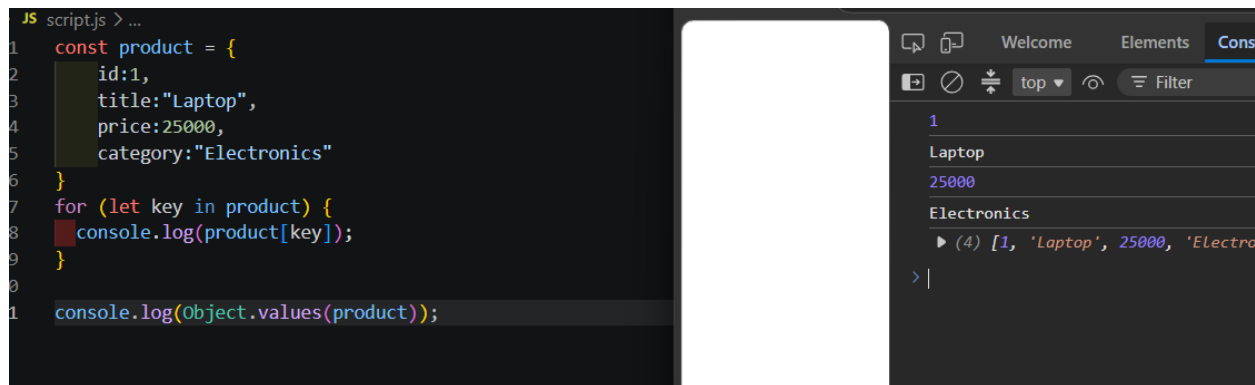
المطلوب

1. باستخدام for...in Keys اطبع كل الـ



2

Values. اطبع كل الـ

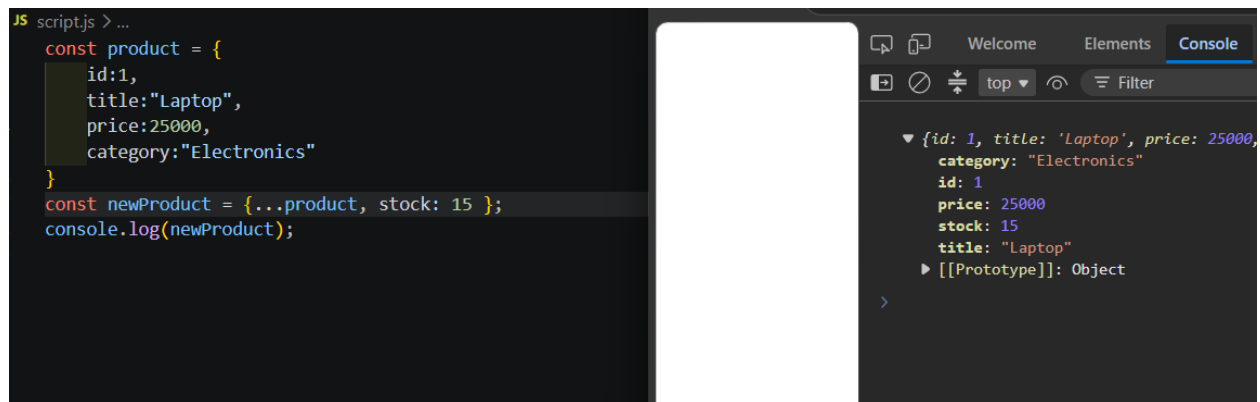


3

جديد وظيف Object اعمل

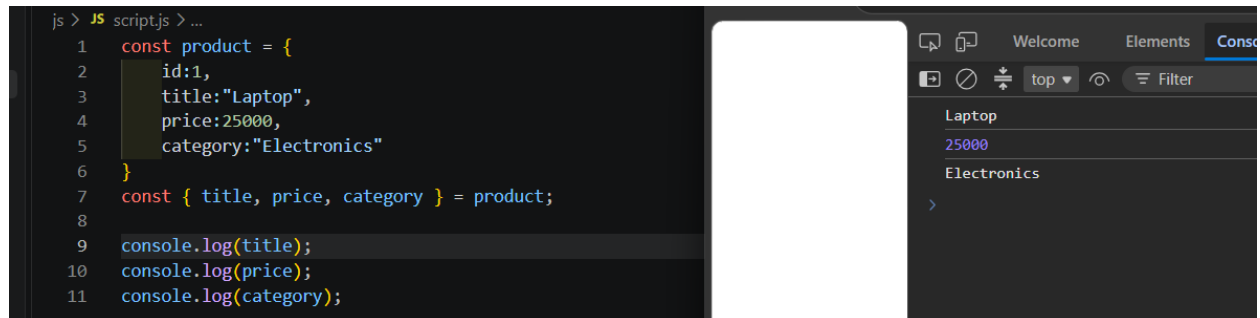
```
```js
```

```
stock:15
```



### 4

استخدم Destructuring.



# Part 9 - Mini Dashboard

اسمها Function اعمل

```js

dashboard()

لما تتنادى تطبع بالشكل ده

Total Employees : 6

Active Employees : ؟

Inactive Employees : ؟

IT Employees : ؟

Highest Salary : ؟

First HR Employee : ؟

Employee Names :

Ahmed

Sara

Ali

```
js > js script.js > dashboard
1  const employees = [
2    { id: 1, name: "Ahmed", age: 22, salary: 6000, department: "IT", active: true },
3    { id: 2, name: "Sara", age: 27, salary: 8500, department: "HR", active: true },
4    { id: 3, name: "Ali", age: 20, salary: 4500, department: "IT", active: false },
5    { id: 4, name: "Mona", age: 30, salary: 10000, department: "Finance", active: true },
6    { id: 5, name: "Omar", age: 24, salary: 7000, department: "Marketing", active: true },
7    { id: 6, name: "Youssef", age: 29, salary: 12000, department: "IT", active: true },
8  ];
9
10 function dashboard() {
11   const total = employees.length;
12   const active = employees.filter(e => e.active).length;
13   const inactive = employees.filter(e => !e.active).length;
14   const itEmployees = employees.filter(e => e.department === "IT").length;
15   const highestSalary = Math.max(...employees.map(e => e.salary));
16   const firstHR = employees.find(e => e.department === "HR");
17   const names = employees.map(e => e.name);
18
19   console.log(`Total Employees : ${total}`);
20   console.log(`Active Employees : ${active}`);
21   console.log(`Inactive Employees : ${inactive}`);
22   console.log(`IT Employees : ${itEmployees}`);
23   console.log(`Highest Salary : ${highestSalary}`);
24   console.log(`First HR Employee : ${firstHR.name}`);
25   console.log(`\nEmployee Names : \n${names.join("\n")}`);
26 }
27
28 dashboard();
```

Click to go back (Alt+Left arrow), hold to see history

127.0.0.1:5500/index.html

Elements

top Filter

Total Employees : 6
Active Employees : 4
Inactive Employees : 2
IT Employees : 3
Highest Salary : 12000
First HR Employee : Sara

Employee Names :
Ahmed
Sara
Ali
Mona
Omar
Youssef

Bonus

reduce() لو تعرف

عادية احسب Loop بدون

1

إجمالي المرتبات

```
js > JS script.js > ...
1 const employees = [
2   { id: 1, name: "Ahmed", age: 22, salary: 6000, department: "IT", active: true },
3   { id: 2, name: "Sara", age: 27, salary: 8500, department: "HR", active: true },
4   { id: 3, name: "Ali", age: 20, salary: 4500, department: "IT", active: false },
5   { id: 4, name: "Mona", age: 30, salary: 10000, department: "Finance", active: true },
6   { id: 5, name: "Omar", age: 24, salary: 7000, department: "Marketing", active: false },
7   { id: 6, name: "Youssef", age: 29, salary: 12000, department: "IT", active: true }
8 ];
9
10 const totalSalaries = employees.reduce((acc, e) => acc + e.salary, 0);
11 console.log(totalSalaries);
```

48000

2

متوسط المرتبات

```
js > JS script.js > ...
1 const employees = [
2   { id: 1, name: "Ahmed", age: 22, salary: 6000, department: "IT", active: true },
3   { id: 2, name: "Sara", age: 27, salary: 8500, department: "HR", active: true },
4   { id: 3, name: "Ali", age: 20, salary: 4500, department: "IT", active: false },
5   { id: 4, name: "Mona", age: 30, salary: 10000, department: "Finance", active: true },
6   { id: 5, name: "Omar", age: 24, salary: 7000, department: "Marketing", active: false },
7   { id: 6, name: "Youssef", age: 29, salary: 12000, department: "IT", active: true }
8 ];
9
10 const avgSalaries = employees.reduce((acc, e) => acc + e.salary, 0) / employees.length;
11 console.log(avgSalaries);
```

8000

3

أعلى مرتب

```
js > JS script.js > ...
1 const employees = [
2   { id: 1, name: "Ahmed", age: 22, salary: 6000, department: "IT", active: true },
3   { id: 2, name: "Sara", age: 27, salary: 8500, department: "HR", active: true },
4   { id: 3, name: "Ali", age: 20, salary: 4500, department: "IT", active: false },
5   { id: 4, name: "Mona", age: 30, salary: 10000, department: "Finance", active: true },
6   { id: 5, name: "Omar", age: 24, salary: 7000, department: "Marketing", active: false },
7   { id: 6, name: "Youssef", age: 29, salary: 12000, department: "IT", active: true }
8 ];
9
10 const highestSalary = employees.reduce((max, e) => e.salary > max ? e.salary : max, 0);
11 console.log(highestSalary);
```

12000

عدد الموظفين النشطين.

```
const employees = [
  { id: 1, name: "Ahmed", age: 22, salary: 6000, department: "IT", active: true },
  { id: 2, name: "Sara", age: 27, salary: 8500, department: "HR", active: true },
  { id: 3, name: "Ali", age: 20, salary: 4500, department: "IT", active: false },
  { id: 4, name: "Mona", age: 30, salary: 10000, department: "Finance", active: true },
  { id: 5, name: "Omar", age: 24, salary: 7000, department: "Marketing", active: false },
  { id: 6, name: "Youssef", age: 29, salary: 12000, department: "IT", active: true }
];

const activeCount = employees.reduce((count, e) => e.active ? count + 1 : count, 0);

console.log(activeCount); // 4
```

